

EXERCISE 10

CRYPTOSYSTEMS BASED ON THE DISCRETE LOGARITHM PROBLEM

1. Understanding groups, cyclic groups and subgroups is important for the use of public-key cryptosystems based on the discrete logarithm problem. That's why we are going to practice some arithmetic in such structures in this and the next few problems.

Let's start with an easy one. Determine the order of all elements of the multiplicative groups of:

- (1) $\mathbb{Z}_5^*$
- (2) $\mathbb{Z}_7^*$
- (3) $\mathbb{Z}_{13}^*$

Create a list with two columns for every group, where each row contains an element a and the order $\text{ord}(a)$.

(Hint: In order to get familiar with cyclic groups and their properties, it is a good idea to compute all orders "by hand", i.e., use only a pocket calculator. If you want to refresh your mental arithmetic skills, try not to use a calculator whenever possible, in particular for the first two groups.)

2. We consider the group $\mathbb{Z}_{53}^*$. What are the possible element orders? How many elements exist for each order?

3. We now study the groups from Problem 1.

- (1) How many elements does each of the multiplicative groups have?
- (2) Do all orders from above divide the number of elements in the corresponding multiplicative group?
- (3) Which of the elements from Problem 1 are primitive elements?
- (4) Verify for each group that the number of primitive elements is given by $\varphi(|\mathbb{Z}_p^*|)$.

4. In this exercise we want to identify primitive elements (generators) of a multiplicative group since they are important for the DHKE and many other public-key schemes based on the DL problem. You are given a prime $p = 4969$ and the corresponding multiplicative group $\mathbb{Z}_{4969}^*$.

- (1) Determine how many generators exist in $\mathbb{Z}_{4969}^*$.
- (2) What is the probability of a randomly chosen element $a \in \mathbb{Z}_{4969}^*$ being a generator?
- (3) Determine the smallest generator $a \in \mathbb{Z}_{4969}^*$ with $a > 1000$.

Hint: The identification can be done naïvely through testing all possible factors of the group cardinality $p-1$, or more efficiently by checking that $a^{(p-1)/q_i} \neq 1 \pmod p$ for all prime factors q_i with $p-1 = \prod q_i^{e_i}$. You can simply start with $a = 1001$ and repeat these steps until you find a generator of $\mathbb{Z}_{4969}^*$.

5. Compute the two public keys and the joint key k_{AB} for the DHKE scheme with the parameters $p = 467$, $\alpha = 2$, and

- (1) $a = 3, b = 5$
- (2) $a = 400, b = 134$
- (3) $a = 228, b = 57$

In all cases, perform the computation of the joint key for Alice and Bob. This is also a perfect check of your results.

6. We now design another DHKE scheme with the same prime $p = 467$ as in Problem 5. This time, however, we use the element $\alpha = 4$. The element 4 has order 233 and thus generates a subgroup with 233 elements. Compute k_{AB} for

- (1) $a = 400, b = 134$
- (2) $a = 167, b = 134$

Why are the session keys identical?

7. In the DHKE protocol, the private keys are chosen from the set

$$\{2, \dots, p-2\}$$

Why are the values 1 and $p-1$ excluded? Describe the weakness of these two values.

8. Given is a DHKE algorithm. The modulus p has 1024 bits and α is a generator of a subgroup where $\text{ord}(\alpha) \approx 2^{160}$.

- (1) What is the maximum value that the private keys should have?
- (2) How long does the computation of the session key take on average if one modular multiplication takes $700 \mu\text{s}$ and one modular squaring $400 \mu\text{s}$? Assume that the public keys have already been computed.
- (3) One well-known acceleration technique for discrete logarithm systems uses short primitive elements. We assume now that α is such a short element (e.g., a 16-bit integer). Assume that modular multiplication with α takes now only $30 \mu\text{s}$. How long does the computation of the public key take now? Why is the time for one modular squaring still the same as above if we apply the square-and-multiply algorithm?

9. This problem demonstrates what can go wrong if a generator is chosen that has certain undesirable properties.

- (1) Show that the order of an element $a \in \mathbb{Z}_p^*$ with $a = p-1$ is always 2.
- (2) What subgroup is generated by a ?
- (3) Briefly describe a simple attack on the DHKE which exploits this property.

10. We consider a DHKE protocol over Galois fields $GF(2^m)$. All arithmetic is done in $GF(2^5)$, with $P(x) = x^5 + x^2 + 1$ as the irreducible field polynomial. The primitive element for the Diffie–Hellman scheme is $\alpha = x^2$. The private keys are $a = 3$ and $b = 12$. What is the session key k_{AB} ?

11. In this chapter we saw that the Diffie–Hellman protocol is as secure as the Diffie–Hellman problem, which is probably as hard as the discrete logarithm problem in the group $\mathbb{Z}_p^*$. However, this only holds for passive attacks, i.e., if Oscar is only capable of eavesdropping. If Oscar can manipulate messages between Alice and Bob, the key agreement protocol can easily be broken! Develop an active attack against the Diffie–Hellman key agreement protocol with Oscar being the man in the middle.

12. Write a program which computes the discrete logarithm in $\mathbb{Z}_p^*$ by exhaustive search. The input parameters for your program are p , α , β . The program computes x where $\beta = \alpha^x \bmod p$.

Compute the solution to $\log_{106} 12375$ in $\mathbb{Z}_{24691}^*$.

13. Encrypt the following messages with the Elgamal cryptosystem ($p = 467$ and $\alpha = 2$):

- (1) $k_{pr} = d = 105, i = 213, x = 33$
- (2) $k_{pr} = d = 105, i = 123, x = 33$
- (3) $k_{pr} = d = 300, i = 45, x = 248$
- (4) $k_{pr} = d = 300, i = 47, x = 248$

Now decrypt every ciphertext and show all steps.

14. Assume Bob sends an Elgamal-encrypted message to Alice consisting of two pieces of plaintext. Since Bob is lazy, he applies the scheme incorrectly and uses the same parameter i for all messages. Assume we know that each of Bob's plaintexts starts with the number $x_1 = 21$, which happens to be Bob's ID. We now obtain the following ciphertexts:

$$(k_{E,1} = 6, y_1 = 12)$$

$$(k_{E,2} = 6, y_2 = 14)$$

The Elgamal parameters are $p = 31, \alpha = 3, \beta = 18$. Determine the second plaintext x_2 .

15. Given is an Elgamal cryptosystem. Bob tries to be especially smart and chooses the following pseudorandom generator to compute new i values:

$$i_j = i_{j-1} + f(j), \quad 1 \leq j$$

where $f(j)$ is a “complicated” but known pseudorandom function (for instance, $f(j)$ could be a cryptographic hash function such as SHA-2) and i_0 is a true random number that is not known to Oscar.

Bob encrypts n messages x_j as follows:

$$k_{Ej} \equiv \alpha^{i_j} \bmod p$$

$$y_j \equiv x_j \cdot \beta^{i_j} \bmod p$$

where $1 \leq j \leq n$. Assume that the last plaintext x_n and all ciphertexts are known to Oscar.

Provide a formula with which Oscar can compute any of the messages $x_j, 1 \leq j \leq n-1$. Of course, following Kerckhoffs' principle, Oscar knows the construction method shown above, including the function $f()$.

16. Given are an Elgamal encryption scheme with public parameters $k_{\text{pub}} = (p, \alpha, \beta)$ and an unknown private key $k_{\text{pr}} = d$. Due to an erroneous implementation of the random number generator of the encrypting party, the following relation holds for two temporary keys:

$$k_{M,j+1} \equiv k_{M,j}^2 \pmod{p}$$

Given are n consecutive ciphertexts

$$(k_{E1}, y_1), (k_{E2}, y_2), \dots, (k_{En}, y_n)$$

belonging to the plaintexts

$$x_1, x_2, \dots, x_n$$

Furthermore, the first plaintext x_1 is known (e.g., header information).

- (1) Describe how an attacker can compute the plaintexts $x_1, x_2, \dots, x_n$ from the given variables.
- (2) Can an attacker compute the private key d from the given information? Explain your answer.

17. Considering the four examples from Problem 13, we see that the Elgamal scheme is nondeterministic: A given plaintext x has many valid ciphertexts, e.g., both $x = 33$ and $x = 248$ have the same ciphertext in the problem.

- (1) Why is the Elgamal encryption scheme nondeterministic?
- (2) How many valid ciphertexts exist for each message x (general expression)? How many are there for the system in Problem 13 (numerical answer)?
- (3) Is the schoolbook RSA cryptographic system nondeterministic once the public key has been chosen?

18. We investigate the weaknesses that arise in Elgamal encryption if a public key of small order is used. We look at the following example. Assume Bob uses the group $\mathbb{Z}_{29}^*$ with the primitive element $\alpha = 2$. His public key is $\beta = 28$.

- (1) What is the order of the public key?
- (2) Which masking keys k_M are possible?
- (3) Alice encrypts a text message. Every character is encoded according to the simple rule $a \rightarrow 0, \dots, z \rightarrow 25$. Three additional ciphertext symbols are the umlauts: $\ddot{a} \rightarrow 26, \ddot{o} \rightarrow 27, \ddot{u} \rightarrow 28$. She transmits the following 11 ciphertexts (k_E, y) :

$$(3, 15), (19, 14), (6, 15), (1, 4), (22, 13), (4, 7), \\ (13, 4), (3, 21), (18, 17), (26, 25), (7, 17)$$

Decrypt the message without computing Bob's private key. Just look at the ciphertext and use the fact that there are only very few masking keys and a bit of guesswork.

19. In this problem, we will introduce the Pohlig–Hellmann exponentiation cipher. It was introduced in 1978 and is a symmetric encryption algorithm based on modular exponentiations, an operation we know from RSA and the DHKE. The cipher is resistant against known-plaintext attacks. From a practical perspective, it has never gained importance since it is far less efficient than block ciphers like AES or stream ciphers. From a pedagogical point of view, however, the Pohlig–Hellman cipher is quite valuable since it combines the discrete logarithm problem with a symmetric cipher without the need to explain the more abstract concept of public-key cryptography.

The Pohlig–Hellmann exponentiation cipher scheme works as follows: Given a (sufficiently large) prime p and a key e , Alice and Bob perform:

- Encryption: $y \equiv x^e \pmod{p}$
 - Decryption: $x \equiv y^d \pmod{p}$, where $d \equiv e^{-1} \pmod{p-1}$
- (1) Compute the encryption of $x = \{S, Y, M, M, E, T, R, I, C\}$ with $p = 29$ and $e = 9$. You can assume a simple mapping of letters to numbers, e.g., $A = 0, B = 1, \dots, Z = 25$.
 - (2) Compute the corresponding key d for decryption and show that the decryption of y yields the plaintext x .
 - (3) Do encryption and decryption work for any e ?
 - (4) Is the scheme still secure if Oscar knows the encryption key e and the prime p ?
 - (5) Can Oscar attack the scheme if he gains access to a pair of plaintext and ciphertext without knowing the key e ?

20. Solving the (generalized) discrete logarithm problem is not hard in all groups. Let us now consider the additive group of integers modulo a prime $\mathbb{Z}_p, (+)$.

- (1) What is the definition of the generalized discrete logarithm problem for this particular group? (cf. Definition 8.3.2)
- (2) Describe a method to efficiently solve the generalized discrete logarithm problem in the group $\mathbb{Z}_p, (+)$.

21. For the DHKE, we choose the generator α as well as the private keys $k_{\text{pr},A}$ and $k_{\text{pr},B}$, which are elements in $\{2, 3, \dots, p-2\}$.

- (1) Show that we cannot choose $p-1$ as a private key in general. Compute $\alpha^{p-1} \pmod{p}$ for any arbitrary DHKE parameters α, p .
- (2) What is the order of the element $\alpha = p-1$? Which subgroup is generated from $\alpha = p-1$?

8.22. Compute the following discrete logarithm in $\mathbb{Z}_{113}^*$ with the Baby-Step Giant-Step method. Justify your answer if a computation is not possible.

- (1) $\alpha = 10, \beta = 98$
- (2) $\alpha = 10, \beta = 99$
- (3) $\alpha = 7, \beta = 98$